# THERMOTRACE AI // SOVEREIGN THERMAL INTELLIGENCE

## OFFICIAL INDUSTRIAL SOURCE INCIDENT DOSSIER

**EVENT REF: EVT-IN-ODI-0018**

National Technical Research Organisation (NTRO) • MoEFCC Oversight

2026-09-02 12:02 IST (2026-09-02 06:32 UTC)

Thermal anomaly **EVT-IN-ODI-0018** is classified as **IND_FIRE** with **96.5% calibrated model confidence** and anomaly severity **CRITICAL**. Peak radiative output reached **720.00 MW** across **1** satellite detection(s). Centroid is associated with **Jindal Steel & Power Angul**.

| | | | |
|---|---|---|---|
| 720.0 MW<br>PEAK RADIANCE | 1 Passes<br>SATELLITE DETECTIONS | 96.5%<br>SOFTMAX CONFIDENCE | CRITICAL<br>ANOMALY SEVERITY |

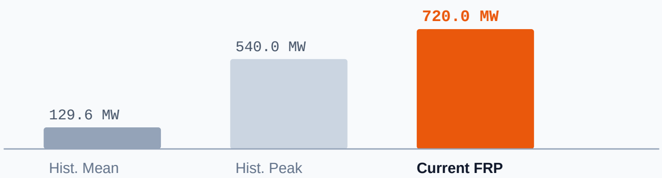
### 1. Verified Satellite Radiometric Metrics

| Metric Name | Observed Value | Historical 90d Baseline | Radiometric Assessment |
|---|---|---|---|
| **Peak Radiative Power (FRP)** | 720.00 MW | Regional Baseline Applied | CRITICAL |
| **Mean Observed FRP** | 510.00 MW | 0 prior events in 90d | Persistence: TRANSIENT |
| **Max Brightness Temp** | 360.0 K | Calibrated Sensor Channels (I4/M13) | Infrared Radiance Confirmed |
| **Detection Temporal Window** | 2026-09-02T03:39 to 05:39 UTC | 1 satellite pass(es) | Trend: STABLE |

### 2. Visual Radiometric Analytics & ML Attribution

**RADIOMETRIC POWER vs HISTORICAL BASELINE**

Z-Score: +3.8σ




**CALIBRATED ML SOURCE PROBABILITIES (XGBOOST 2.0)**


**ESA WORLDCOVER 10m SPATIAL TERRAIN COMPOSITION (3.5 km RADIUS)**

## 3. Centroid Location & Infrastructure Context

| Attribute | Registered Forensic Detail | Attribute | Registered Forensic Detail |
|---|---|---|---|
| **Centroid Coordinates** | **20.83300°N, 85.15100°E** | **Primary Land Use** | Unknown |
| **State / Territory** | Odisha | **District / Jurisdiction** | - |
| **Associated Facility** | **Jindal Steel & Power Angul** | **Distance to Boundary** | **0.12 km** |
| **Industrial Sector** | Steel Plant | **Facility Operator** | Verified Entity |

## 4. Nearby Industrial Infrastructure (3.5 km Radius)

| Facility Name | Sector & Facility Type | Distance from Centroid | Attribution Audit |
|---|---|---|---|
| **Jindal Steel & Power Angul** | Steel Plant | **0.35 km** | Within Proximity Buffer |
| **Industrial Facility** | Coal Mine • Industrial Unit | **3.32 km** | Within Proximity Buffer |
| **Industrial Facility** | Coal Mine • Industrial Unit | **10.81 km** | Outside Primary Buffer |
| **Industrial Facility** | Coal Mine • Industrial Unit | **12.32 km** | Outside Primary Buffer |

## 5. Multi-Sensor Satellite Radiometric Passes

| # | Detection Time (UTC) | Sensor Platform | Radiant Output | Brightness (K) | Track Quality |
|---|---|---|---|---|---|
| 1 | **2026-09-02T03:39** | **VIIRS 375m** | **720.00 MW** | **360.0 K** | Direct Telemetry Ingest |

## 6. Recommended Forensic Follow-Up & Compliance Actions

**ACTION 01:** Compare current FRP against the recent facility operating baseline.

**ACTION 02:** Review repeated activity across recent satellite passes.

**ACTION 03:** Verify whether activity is consistent with known facility operations.